Predictive ML trading assistant for a first-time crypto trader (BTC/USDT)

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In the interview you said the end result should be simple enough for someone who doesn't know much about finance, so I designed it around that person. Picture someone with $1,000 set aside for crypto who asks, "Bitcoin is up this week, should I buy?" A price forecast won't help them much. What they need to know is how much of that money should be in Bitcoin right now, and, if they buy today, what usually happens next and what they could lose. The assistant answers those two questions in plain words: the first once a week, the second every four hours.

1. What it predicts, and why that helps. The weekly answer comes from five trend checks: is Bitcoin's price higher than it was 1, 2, 4, 8 and 16 weeks ago? Each yes means holding one fifth of the Bitcoin budget, so 3 out of 5 means 60% in Bitcoin and 40% in cash. This is a simple prediction that recent direction tends to continue, the time-series momentum effect Liu and Tsyvinski (2021) found in crypto. I average five lookbacks so I never had to pick a "best" one.

The four-hourly answer is the machine-learning part. Take a specific trade: buy at the next candle's open, set a stop 1 unit of expected daily movement below and a target 1.5 units above, and close it after 24 hours if neither is hit. The model predicts which of the three happens first: target, stop, or sold after 24 hours. That last case can end up or down, and in this data it's the most common outcome (about 56% of trades). The expected daily movement is an exponentially weighted average of past 4-hour returns with a 3-day half-life, scaled to 24 hours. Both levels stay fixed once the trade is open. If one candle touches both levels I count it as a stop, which only happened in 7 of 18,396 cases. If the price gaps below the stop, the exit is at that worse price. I chose this target because, for a beginner, the answer already includes the stop. Most beginners don't lose because their forecast was slightly off; they lose because nothing limited the loss.

2. Data and features. Binance BTCUSDT 4-hour candles, January 2018 to June 2026 (18,510 after cleaning), plus the 1-hour file for cross-checking. I rebuilt every 4-hour candle from the 1-hour data and 18,473 of 18,474 matched within 0.2%. The mismatch check found two corrupted rows on 6 July 2018 showing Bitcoin at $108,000 when it traded near $6,500, so I removed them. I didn't fill gaps: 16 candles are missing (exchange outages before 2020), and I dropped the 48 trades whose 24 hours crossed one of them. The short-term model uses 18 inputs a trader would recognise: recent returns from 4 hours to 7 days, how jumpy the market is, where price sits against its 7- and 30-day highs and lows, the 8-day trend and how far price has run from it, volume compared with normal, the share of trading done by eager buyers, and time of day. Every input uses only data available at that moment. A test checks this by deleting every later candle and confirming nothing changes. Funding rates and order-book data aren't in this dataset, so they're on the later list.

3. Models and baselines. The trend gauge has nothing fitted: five checks with equal weight. With eight years of one asset, fitted weights would mostly fit noise (the argument DeMiguel, Garlappi and Uppal make for equal weights). For the short-term trade I fixed a rule before seeing results: start with a 5-rule points scorecard and only switch to a bigger model if it beats the scorecard on both ranking quality and profit over 2022 to mid-2025. The challengers were a

regularised multinomial logistic regression and a shallow boosted-tree model. On real data the scorecard found nothing and the logistic regression won. Its probabilities are recalibrated on the latest 20% of each training window, kept separate from fitting. The baselines are buy and hold; cash; "take every trade" with the same stop, target and costs (this shows whether the ML adds anything beyond the trade rules); and the same number of trades placed at random times.

4. How I tested it. Everything is split by time. The short-term model trains from 2018 and is tested a quarter at a time from January 2022, refitting each quarter. The last 12 months (15 June 2025 to 15 June 2026) were kept as a final test. Trades can overlap in time, so before each test period I remove training trades whose 24 hours hadn't finished, plus a one-day buffer. Costs are 0.10% fee plus 0.03% slippage each way, and I check the trading results again at double those costs. It's long only, one trade at a time, no leverage, and I report time in the market next to returns, because a strategy can look safer just by being invested less. Every variant I tried is logged, and I report the Deflated Sharpe Ratio (Bailey and López de Prado 2014), a block bootstrap against buy and hold, and a test against random entries. There are 18 automated tests, including a trade worked out by hand (buy at 100, exit at 103, +2.73% after costs). One honest note: I added the trend gauge after I'd seen the test year, so I also report it on 2018 to 2021, which I never used for any decision, and I didn't tune anything.

Results. The trend gauge, checked only on Sundays (about 37 changes a year), after costs:

Period	Buy and hold	Trend gauge	Time in Bitcoin
2018 to 2021	+517%, worst fall −63%	+749%, worst fall −40%	57%
2022 to mid-2025	+128%, worst fall −67%	+109%, worst fall −34%	53%
Test year (Jun 2025 to Jun 2026)	−38%, worst fall −51%	−15%, worst fall −25%	43%
All of 2018 to 2026	+778%, worst fall −77%	+1,408%, worst fall −47%	53%

Over 2018 to 2026 that's about 40% a year against 31%, with the worst fall cut from −77% to −47%. The smaller falls show up in every period and for every lookback, and that's the part I'm confident about. The higher Sharpe ratio (1.06 vs 0.75) isn't statistically significant: the bootstrap range is -0.09 to +0.70. From 2022 to mid-2025 it made slightly less than holding.

The short-term model was weaker. From 2022 to mid-2025 it took 182 trades, averaged +0.10 times the amount risked per trade, and beat all 500 random-entry versions. Sized so each stop costs about 1% of the account, that's +17% with a worst fall of −7%. But it fails my release test: at least 100 trades, a 95% lower bound on average profit above zero, still profitable at double costs, and better than taking every trade. The lower bound is -0.02, and at double costs it slightly loses. In the test year it took 5 trades and lost 3% with 1% sizing. So the app shows it in research mode: you can see the odds and the risk, but it never says buy.

5. What the trader sees. One screen, in plain words:

- The main answer, e.g. "Hold 60% of your Bitcoin budget this week", with the five trend boxes as up or down arrows. If data is missing it says "Data unavailable, no signal" instead of guessing.

- Today's trade check, labelled "Research only" until it passes the release test, with odds shown as 10 icons (target, sold after 24 hours, stop), because "3 out of 10" is far easier to read than "30%" (Gigerenzer and Hoffrage 1995; Galesic et al. 2009).
- The trade in dollars for a $1,000 account, for example: buy $360 of Bitcoin; if the stop is hit you lose about $10 including fees; if the target is hit you gain about $12.66 after fees; otherwise sell after 24 hours, up or down. A sudden price gap can cost more. Each check shows when it expires.
- The two or three biggest factors behind the model's lean, labelled as what the model is reacting to rather than reasons the price will move, and a full record of past signals, losses included, that you can replay.

6. Risks, limits and next steps. The biggest risk is that there's no stable short-term edge after costs, and my results suggest that's the case today. Trend following lags in choppy, sideways markets and checks only weekly, so it can't dodge a one-day crash like March 2020. Stops don't help in a gap, eight years of one coin is limited data, Bitcoin is a survivor so this may not carry over to smaller coins, and exchange or stablecoin problems are outside the model. The user is a risk too, by moving stops or trading too often, so risk is fixed at 1% and "stay out" is the default. Next I'd paper-trade both parts for a couple of months and compare them with the backtest each month, as I did for live-versus-backtest monitoring at Oakfield. I'd also add funding data and try ETH.

Where this comes from. The time splits, purging and Deflated Sharpe come from my Inter IIT crypto bootcamp project (BTC/ETH with PPO, XGBoost and GARCH), the 1% risk cap from my NIFTY 100 intraday framework, costs inside the decision from Kriti 2026, rank-IC checks from my NIFTY 100 factor work, and keeping the model simple from the Ebullient challenge, where my 60+ feature PPO agent worked but was hard to explain.

References: Bailey and López de Prado (2014), J. Portfolio Management. DeMiguel, Garlappi and Uppal (2009), Review of Financial Studies. Galesic et al. (2009), Health Psychology. Gigerenzer and Hoffrage (1995), Psychological Review. Liu and Tsyvinski (2021), Review of Financial Studies. López de Prado (2018), Advances in Financial Machine Learning.